

COMPUTER SCIENCE TRIPOS Part IB – 2024 – Paper 4

1 Compiler Construction (jdy22)

Here are two grammars for languages built from terminals **a** and **b**:

Grammar A:

$S \rightarrow C a C b A$
 $A \rightarrow B$
 $A \rightarrow S$
 $B \rightarrow \epsilon$
 $B \rightarrow b$
 $C \rightarrow \epsilon$

Grammar B:

$S \rightarrow A b$
 $A \rightarrow \epsilon$
 $A \rightarrow A a b$

In both grammars **S** is the start symbol.

context-free
grammars

(a) The languages $L(A)$ and $L(B)$ for grammars A and B are not equal.

(i) Give an example of a string that is in $L(A)$ but not in $L(B)$.

(ii) Give an example of a string that is in $L(B)$ but not in $L(A)$.

[2 marks]

Answer:

(i) **ab**

(ii) **b**

regular
expressions,
context-free
grammars

(b) Give regular expressions describing each of $L(A)$ and $L(B)$.

[4 marks]

Answer: For $L(A)$: **$ab(ab)^*(b|\epsilon)$** . (Or, in extended notation: **$(ab)^+b?$** .) For $L(B)$: **$(ab)^*b$** .

top-down parsing

(c) Compute NULLABLE, FIRST sets and FOLLOW sets for each grammar.

[6 marks]

Answer:

Grammar A:

	S	A	B	C
NULLABLE	$\perp$	$\top$	$\top$	$\top$
FIRST	{a}	{a,b,ϵ}	{b,ϵ}	{ϵ}
FOLLOW	{$\\$}	{$\\$}	{$\\$}	{a,b}

Grammar B:

	S	A
NULLABLE	$\perp$	$\top$
FIRST	{a,b}	{a,ϵ}
FOLLOW	{$\\$}	{a,b}

top-down and
bottom-up
parsing

(d) State with justification whether each grammar is LL(1) and whether it is SLR(1).

[8 marks]

Answer:

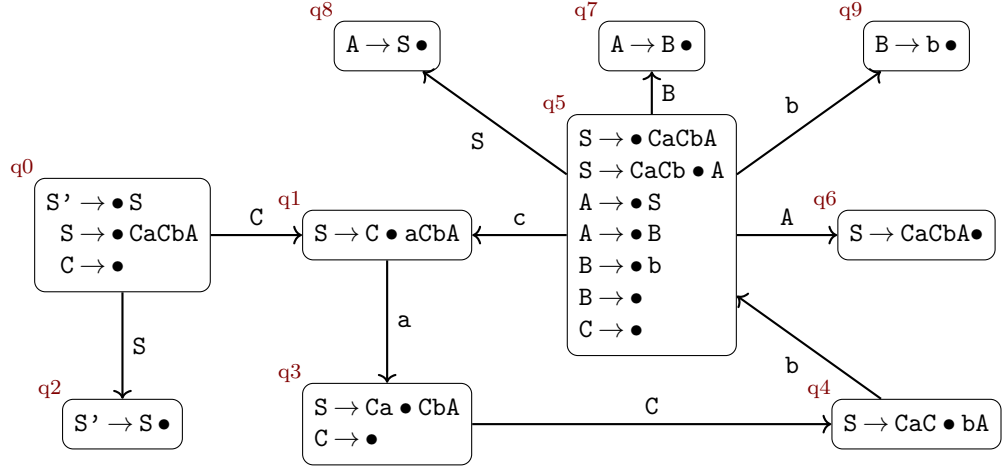
(i) Grammar A is LL(1), but it is not SLR(1).

— Solution notes —

To see that it is LL(1), construct the table:

	a	b	\$
A	S	B	B
B		b	ϵ
C	ϵ	ϵ	
S	C a C b A		

To see that it is not SLR(1), construct the automaton:



and observe that there is a shift/reduce conflict for token **b** in state q_5 :

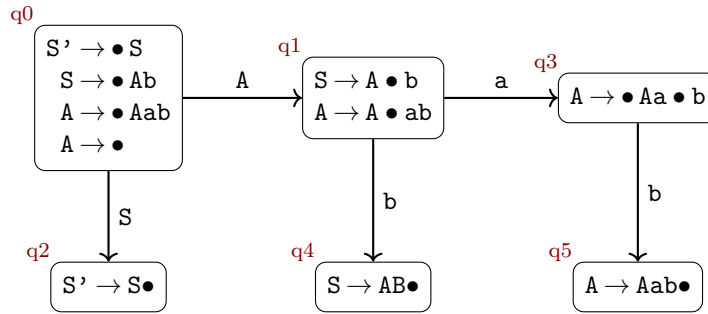
Since $B \rightarrow \bullet b \in q_5$ and $\delta(q_5, b) = q_9$, we can shift to q_9 .

Since $C \rightarrow \bullet \in q_5$ and $b \in \text{FOLLOW}(C)$, we can reduce to q_4 using $C \rightarrow \epsilon$.

(ii) Grammar B is not LL(1), but it is SLR(1).

It is easy to see that it is not LL(1) because A is left recursive.

To see that it is SLR(1), first construct the automaton:



and then construct the action table

	a	b	\$
q0	r2	r2	
q1	s3	s4	
q2			acc
q3		s5	
q4		r1	
q5	r3	r3	

and observe that there are no conflicts.